From interaction history to action

GRU state estimation with a fixed physical model

Current observation: $x_t,\ v_t,\ F_t$

Candidate commands $u^{(j)}$ and controller K, D

Interaction history

$H_t = \{x, v, F, u, \Delta t\}_{t-H:t}$

3 seconds / 301 samples

GRU encoder

Recurrent memory

h_t (32 dimensions)

State estimate

Head + physical constraint

$\hat{z}_t = (\hat{z}_{1,t}, \hat{z}_{2,t})$

Physics rollout

Fixed material law

$\hat{s}_0 = (x_t, v_t, \hat{z}_t)$

$\hat{x}, \hat{F}$

Action selection

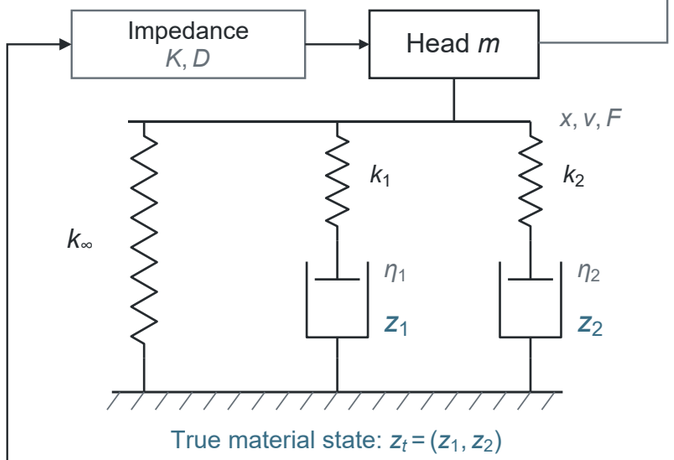
Respect the force limit

Minimize completion time

Measurements x, v, F

Simulated interaction

1D contact / two material states



True material state: $z_t = (z_1, z_2)$

Inside each physics rollout

Every candidate starts from the same estimated state.

Initial state		Predicted state		Predicted state
$\hat{s}_0$	f_{phys}	$\hat{s}_1$	f_{phys}	$\hat{s}_2$
$x_t,\ v_t$		$\hat{x}_1,\ \hat{v}_1$		$\hat{x}_2,\ \hat{v}_2$
$\hat{z}_{1,t},\ \hat{z}_{2,t}$	$u_0^{(j)}$	$\hat{z}_{1,t+1},\ \hat{z}_{2,t+1}$	$u_1^{(j)}$	$\hat{z}_{1,t+2},\ \hat{z}_{2,t+2}$

Material-state components are highlighted in blue.

Execute selected command u^*

One decision at time t: infer state, compare candidate futures, then execute.